Probing Functional Plasticity in Peptide–Protein Binding with Minimal Data

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Abstract

Machine learning has accelerated protein design, yet peptide discovery remains constrained by scarce training data. We present Minimal Data · Maximal Insight (MDMI), an ML-assisted structure-aware pipeline that integrates AlphaFold-Multimer predictions, knowledge- and physics-based scoring, and a compact ML model to guide peptide optimization. Using split GFP as a testbed, MDMI generated functional peptide binders with up to 63% sequence divergence from the wild type. When benchmarked against RFdiffusion and PepMLM, MDMI produced fourfold more high-confidence binders while requiring nearly 100 labeled examples. Through discovering novel functional variants with high sequence divergence, MDMI pipeline demonstrates capability to maps functional plasticity in peptide–protein interfaces, offering a practical framework for low-N peptide engineering.

1 Introduction

Peptides, typically ranging from 2 to 50 amino acid residues, represent a fundamental class of biomolecules with extraordinary versatility in biological systems. Their significance spans multiple domains, from cellular signaling and hormonal regulation to potential therapeutic interventions and antibacterial mechanisms. Unlike larger protein molecules, peptides offer distinct advantages such as enhanced cellular penetration [Thapa and Sullivan, 2018] and lower immunogenicity [Real-Fernandez et al., 2023]. Given their multifaceted roles and enormous demand for peptide biologics, computational strategies, particularly those based on machine learning (ML), have become important tools for accelerating peptide design and discovery [Goles et al., 2024, Nielsen et al., 2024] with examples such as RFpeptide [Rettie et al., 2024], Pepflow [Abdin and Kim, 2024], PepMLM [Chen et al., 2024], and Peptune [Tang et al., 2025] for *de novo* design. Although ML-guided discovery workflows can be extremely powerful tools, their effectiveness is often hindered by the scarcity of extensive datasets required for training robust models [Alzubaidi et al., 2023].

Recently, techniques such as biophysics-based protein language models [Gelman et al., 2025], deep learning-based [Biswas et al., 2021], and few-shot learning [Zhou et al., 2024], have been developed

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for single proteins to aid in data-scarce scenarios. While these techniques have been successful in addressing data scarcity for single proteins, biologics usually do not work in isolation. Thus, there is a clear need for models that can consider both binder partners, as peptide-protein binding involves complex structural and functional dependencies that single-protein models fail to capture effectively.

Here we describe Minimal Data Maximal Insight (MDMI), an ML-assisted workflow for peptide discovery and diversification which identifies novel binders from a limited dataset through a structural-based approach. In MDMI, we work with approximately 100 well-characterized peptide sequences to a specific target protein. We focus on a split Green Fluorescent Protein (GFP) system with GFP11, a 16-amino acid fragment as our peptide model, and its complementary larger fragment, GFP1-10 (217 amino acids) as our target protein proxy. When GFP1-10 and GFP11 successfully interact, they reconstitute into a functional GFP complex, emitting fluorescence, which serves as an optical readout of functionality (Figure 1a). We used MDMI to find functional alternatives for the wild-type GFP11 peptide.

We first assessed MDMI on a public avGFP brightness dataset [Sarkisyan et al., 2016] in which we identified 109 mutations in the GFP11 region. After validating the predictor on this benchmark, we generated an independent low-N dataset using the split superfolder GFP (sfGFP) system, thereby demonstrating that MDMI remains effective even when no pre-existing training corpus is available. With a single round of experiments, and building on the motifs leading to high-activity variants, we could show that our pipeline is able to explore the functional plasticity of GFP11. Throughout, we use functional plasticity to mean that the peptide–protein interface retains function despite substantial sequence divergence in the peptide, that is, multiple sequence-distinct peptides can bind productively. We obtained GFP11 variants bearing up to 62% mutation relative to the wild type while preserving function.

We compared MDMI against RFdiffusion [Watson et al., 2023], the current state-of-the-art zero-shot generator, as well as PepMLM [Chen et al., 2024], a protein language model-based algorithm for structure-independent peptide binder generation, and observed that, MDMI produced approximately four-fold more high-confidence binders.

By explicitly encoding interactions and by operating effectively with scarce data, MDMI provides a practical route to explore peptide sequence–function landscapes. We anticipate that the framework will prove useful across applications ranging from antimicrobial discovery to targeted drug delivery, wherever limited experimental throughput and interface specificity are the primary bottlenecks.

2 Pipeline

Our pipeline combines a structure-based predictive model with a generative model. In the sequence-agnostic predictive model (Figure 1b.), we began by simulating 3D structures of the GFP1-10/GFP11 complex for each mutant using AlphaFold-Multimer [Evans et al., 2022]. These structures were scored using SPserver [Aguirre-Plans et al., 2021] for statistical potentials and PyRosetta [Chaudhury et al., 2010] for physics-informed evaluation. Next, an ensemble model was trained based on these scores. Together, these components form the foundation for our peptide binder prediction model. Next, a genetic algorithm, implemented via PyGad, introduces genetic diversity and broadens the search space for peptide sequence variations (Figure 1c). Starting with an initial randomized population of sequences, each sequence's functional potential is assessed, focusing on brightness when bound to GFP1-10. Sequences with higher potential are used to generate offspring, combining traits through crossover operations. Top candidates, exhibiting high solubility, are selected for experimental testing.

To test our model, we first drew on a publicly available avGFP mutant dataset providing brightness measurements [Sarkisyan et al., 2016]. From this resource, we filtered 109 variants bearing substitutions exclusively in the GFP11 segment. On the training and hold-out sequences the best performing ensemble achieved: Root Mean Square Error (RMSE) = 3.57 (train), 9.36 (val); accuracy = 100% / 88.9%; Matthews correlation coefficient (MCC) = 1.00 / 0.75; Area Under the Receiver Operating Characteristic Curve (AUROC) = 0.98 / 0.91; Area Under the Precision-Recall Curve (AUPRC) = 0.99 / 0.98. The lower validation R^2 reflects calibration error but the rank ordering of sequences is preserved, as shown by AUROC = 0.91. Because downstream decision-making depends only on whether a candidate exceeds the fluorescence threshold, classification metrics (accuracy = 88.9%, MCC = 0.75) are the most relevant performance indicators.

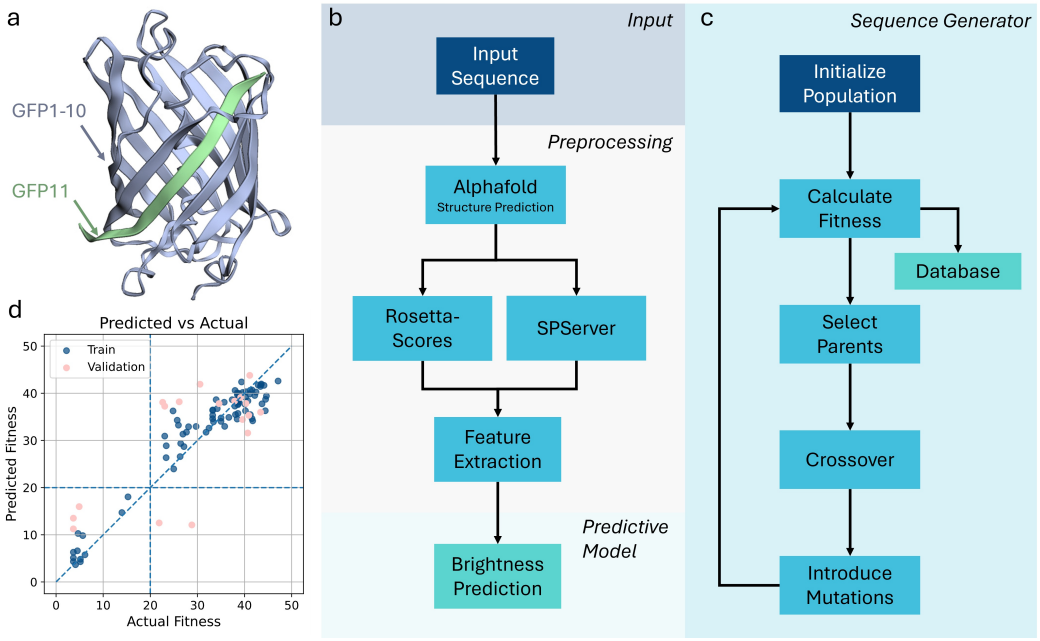
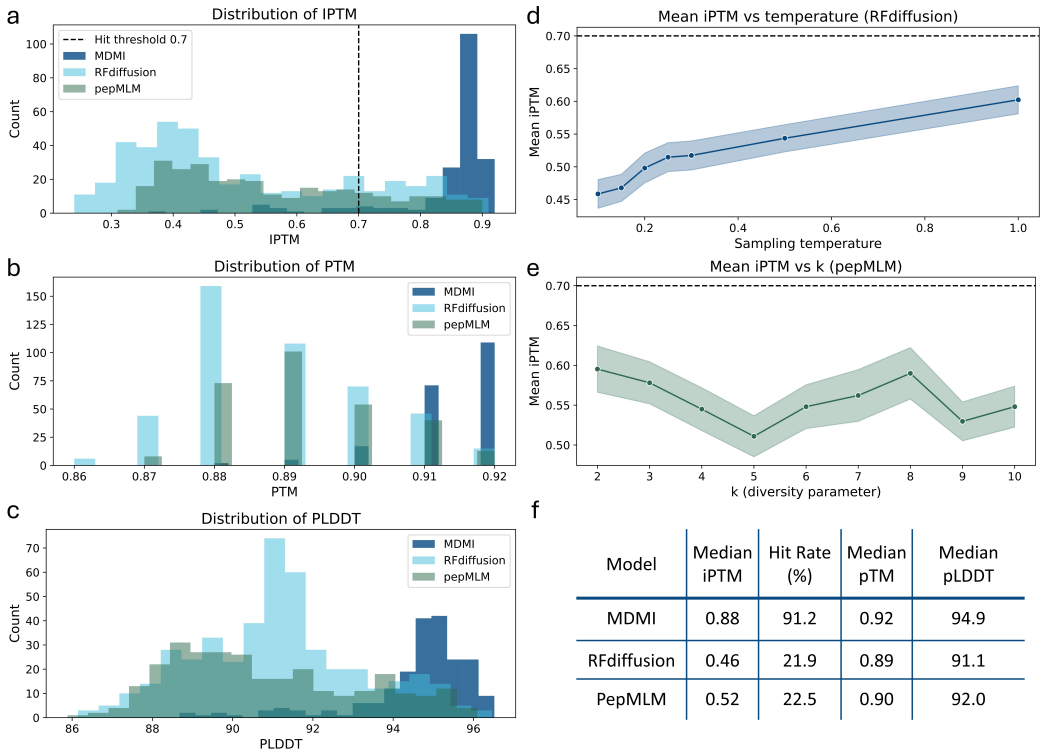


Figure 1: **a)** Structural model of the split GFP system used in this study. The GFP11 peptide (green) binds to the larger GFP1-10 fragment (blue), and the degree of complementation correlates with fluorescence. **b)** Workflow for building the predictive model. Input peptide sequences are folded with AlphaFold-Multimer. Structural features are extracted using Rosetta scoring functions and SPServer metrics. These features are used to train a model predicting peptide-dependent fluorescence. **c)** Sequence generation pipeline. A genetic algorithm initializes a population of candidate sequences, evaluates their predicted fitness using the model, and iteratively improves them through selection, crossover, and mutation. **d)** Performance of the trained model. Predicted versus actual experimental fitness values are shown for both training (blue) and validation (pink) datasets

3 Benchmarking

Building on the model's promising performance on the publicly available dataset, we generated our own experimental dataset to retrain the model and implemented a genetic algorithm to design new sequences. The original dataset, sourced from the literature [Sarkisyan et al., 2016], involved in vivo expression of mutants of avGFP. We hypothesized that a dataset collected under conditions consistent with our in vitro cell-free experiments would enhance the model's performance. Consequently, we conducted a parallel screen of cell-free expression of 114 GFP11 variants and measured their brightness levels relative to the wild-type sequence after mixing with purified GFP1-10 to generate our own dataset.

Using an in-house dataset, we re-trained our model and ran the GA to find novel functional variants of GFP11. We benchmarked MDMI using the metrics reported by alphafold against RFdiffusion, a general-purpose zero-shot backbone generator conditioned on target structure [Watson et al., 2023], as well as PepMLM, a sequence-only span-masked language model trained on 10000 protein-peptide binders [Chen et al., 2024]. We evaluated design quality with AlphaFold-Multimer's predicted template modeling (TM) score metrics (iPTM and pTM) and residue-level confidence (pLDDT) (Figure 2). AlphaFold-Multimer provides three confidence metrics: pLDDT, which reports residue-level structural integrity; pTM, an integrated score for the accuracy of the complex's overall topology; and iPTM, which specifically gauges how reliably the model positions the interacting subunits relative to each other, an indicator of interface credibility and potential binding affinity. The MDMI set (n = 204) showed a median iPTM of 0.88 (inter-quartile range [IQR] 0.86–0.90), whereas the RFdiffusion set (n = 448) had a median of 0.46 (IQR 0.37-0.63). Applying an in silico threshold of iPTM$>$0.7 for effective binding (that is, the predicted interface between the peptide and the target is well formed and likely to be physically realistic), 90.2% of MDMI designs qualified as 'hits' compared to 21.9% for RFdiffusion and 22.5% for PepMLM. Consistent trends were observed for



Model	Median iPTM	Hit Rate (%)	Median pTM	Median pLDDT
MDMI	0.88	91.2	0.92	94.9
RFdiffusion	0.46	21.9	0.89	91.1
PepMLM	0.52	22.5	0.90	92.0

Figure 2: **a)** Distribution of interface TM-scores (iPTM) predicted by AlphaFold-Multimer for MDMI (navy), RFdiffusion (cyan), and pepMLM (forest green) designs. The dashed line marks the in silico success threshold (0.7). **b)** Predicted TM-score (pTM) distributions for the same design sets. **c)** Per-residue backbone confidence (pLDDT) distributions. **d)** RFdiffusion designs generated at increasing sampling temperatures show a modest upward trend in mean iPTM. Shaded band indicates ±1 standard error of the mean (SEM); the dashed line marks the in silico hit threshold (iPTM=0.7). **e)** PepMLM designs sampled with top-k= 2–10 show a non-monotonic relationship between diversity and quality. Mean iPTM dips at k=5, recovers above k=6, and peaks at k=8. Shaded band: ±1 SEM. **f)** Summary statistics for each method. Table reports the median iPTM, hit rate (percentage of designs with iPTM larger than 0.7), median pTM, and median pLDDT.

the backbone metrics: MDMI achieved a median pTM score of 0.92 and a median pLDDT of 94.9, while RFdiffusion reached 0.89 and 91.1, and PepMLM reached 0.90 and 92, respectively.

4 Experimental Validation

Using our homemade dataset, we employed the genetic algorithm to identify novel sequences with 3-6 mutations, corresponding to approximately 19-38% mutation rates. The genetic algorithm evaluated 1327 sequences, from which 40 sequences with the highest predicted functionality and high solubility were selected for wet lab validation. The variatns were expressed using a cell-free system and were mixed with purified GFP1-10. As illustrated in Figure 3a, the sequences within the designed batch exhibited significantly higher signals than the control group. Specifically, 63% of the designed sequences exhibited more than 20% activity relative to the wild-type, 28% showed over 60% activity, and 20% demonstrated greater than 75% activity compared to the wild-type indicating successful enhancement through structural modeling and genetic algorithm as well as employing a well-curated dataset.

Next, to further understand the underlying motifs contributing to successful binding and functionality, we analyzed the sequences that exhibited more than 60% activity. By identifying frequently occurring amino acids within these sequences, we assembled new variants incorporating these residues. This approach led to sequences with 10 amino acid differences, representing significant deviations

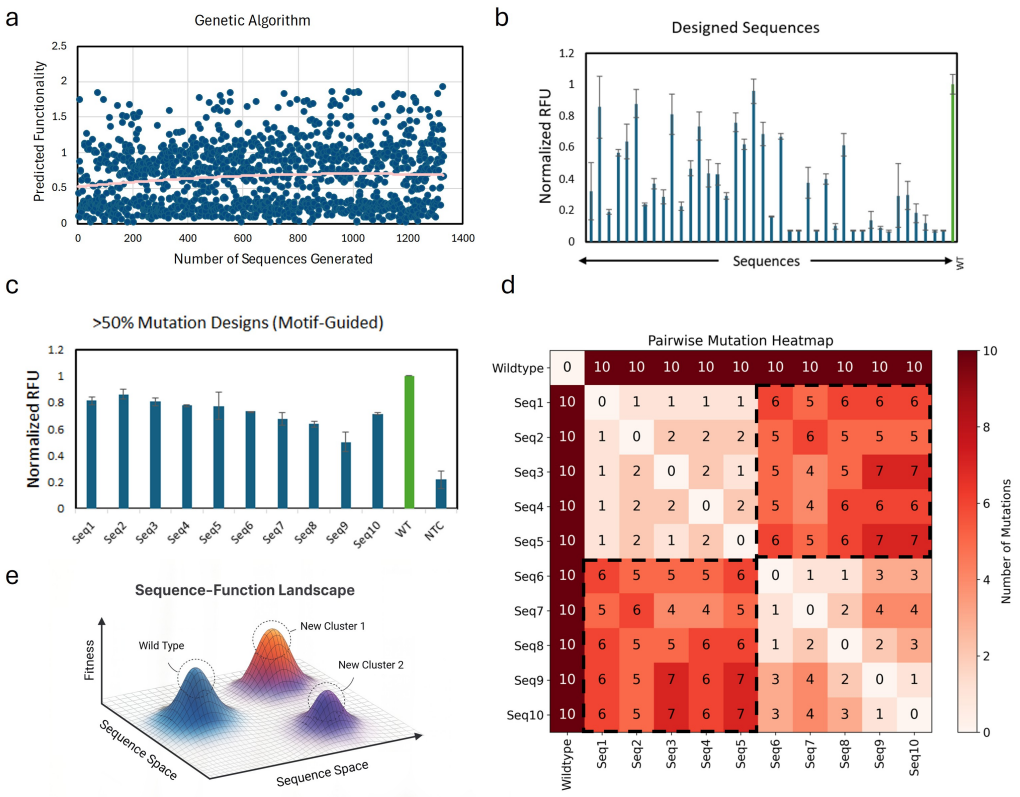


Figure 3: **a)** Predicted functionality of sequences over generations in the genetic algorithm. While most sequences remain near baseline, the model identifies variants with consistently higher predicted functionality. **b)** Experimental fluorescence (normalized RFU) of designed peptides. A range of brightness values is observed, with several sequences approaching wild-type levels. **c)** Sequence divergence expression activity of GFP11 variants **d)** Pairwise amino acid substitution heat-map for the ten variants. Cell values denote the number of mismatches between the sequences indicated on the axes. **e)** Schematic showing the new clusters discovered using MDMI pipeline

from the wild type (62% sequence difference). The results of the screening, shown in Figure 3c, revealed that all of the designed sequences displayed a high degree of fluorescence compared to the wild-type GFP11, despite high sequence divergence.

To probe how far we had traveled through sequence space in a single round of screening, we mapped the ten variants onto a pairwise substitution matrix (Figure 3d). Every variant carries ten substitutions relative to wild type GFP11 (10/16 residues), yet the matrix reveals two discrete clusters. Seq1–Seq5 form a tightly knit group that differ from one another at less than 2 positions, whereas Seq6–Seq10 constitute a second group, again internally close, but separated from the first cluster by more than 4 substitutions. Thus, our motif-guided search accessed two distant hills on the sequence–function landscape (Figure 3e).

Large sequence differences across peptide designs offer an opportunity to discover alternative binding solutions that exploit physicochemical motifs absent from the wild type sequence. In protein–peptide interactions, distinct residue sets can converge on similar three-dimensional contacts through compensatory side-chain rearrangements, ultimately yielding new epitopes with altered stability, specificity, or immunogenicity. Such diversity is desirable in applications that require orthogonal binders, reduced immunogenicity, or freedom to operate around existing intellectual-property claims.

5 Conclusion

Data scarcity has long hampered predictive modeling of short, flexible binders because the combinatorial explosion of admissible sequences quickly outpaces feasible wet-lab throughput. Prior approaches either relied on pretrained language models that assume the target sequence distribution resembles natural proteins, or on large, supervised datasets that are generally unattainable for newly emergent targets. In contrast, MDMI exploits a structure-aware model, allowing it to generalize from a few dozen labeled examples. The success we observe with the GFP split system suggests that high-quality, interaction-aware features can compensate for limited sample size; this principle should translate to other peptide–protein interfaces provided that the complex can be modelled with reasonable confidence.

By decoupling peptide design from large datasets, MDMI democratizes access to advanced computational tools for labs with limited resources. With only one round of screening, we were able to train a machine learning algorithm capable of predicting sequences with high degree of functionality. MDMI's ability to engineer highly divergent yet functional sequences opens doors to *de novo* peptide therapeutics with reduced immunogenicity and enhanced stability. Demonstrated on GFP11, MDMI's pipeline is adaptable to other peptide-protein systems, offering a blueprint for accelerating therapeutic peptide discovery.

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